

TENSOR ANALYSIS

SLIDES WEEK 24 – LECTURE 1

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2025/26

Portfolio

- The **Portfolio Test** will take place on 11 March, 2026.
 - ▶ Duration: 1:15 minutes (+25 minutes if you are entitled to extra time).
 - ▶ Main Cohort: INB2101
 - ▶ PASS or Reasonable adjustments: DCB1105
- The content covers everything we have learned up to the end of this week.

Portfolio overview

Part A — Take-home (25%)

- Questions are already available.
- Complete at your own pace at home.
- You must **bring Part A** with you to the TCA.

Part B — Time Constrained Assessment (TCA) (75%)

- Takes place this week on **11 March 2026**

Submission

- Submit **Part A and Part B together** as a **single PDF file**.
- If you cannot submit both together, **e-mail part A** to me.

CHAPTER 5: TENSORS IN GENERALISED COORDINATE SYSTEMS

Today: Chapter 5– Tensors in generalised coordinate systems

1. Covariant, contravariant and mixed components of a tensor,
2. Associated tensors,
3. The metric tensor,
4. Higher order tensors in generalised coordinates.

RECAP OF PREVIOUS CHAPTERS

Chapter 3

We already considered local coordinate transformations of a rank 1 tensor (vector).

We discussed:

- Method of dual bases,
- Covariant and contravariant components of a vector,
- The transformation rule,
- Relationship between covariant and contravariant components of a vector,
- The metric tensor g_{ik} (or g^{ik}).

RECAP: DUAL BASES

Dual basis

Two bases $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ and $\mathbf{e}^1, \mathbf{e}^2, \mathbf{e}^3$ are *dual* if they satisfy

$$\mathbf{e}_i \cdot \mathbf{e}^k = \begin{cases} 0 & \text{if } i \neq k, \\ 1 & \text{if } i = k. \end{cases}$$

The dual basis $\mathbf{e}^i$ can be found from the original basis $\mathbf{e}_i$ via

$$\mathbf{e}^i = \frac{\mathbf{e}_j \times \mathbf{e}_k}{\mathbf{e}_i \cdot (\mathbf{e}_j \times \mathbf{e}_k)}.$$

RECAP: COVARIANT AND CONTRAVARIANT COMPONENTS

Covariant and contravariant components

Using dual bases, we have found that vector $\mathbf{A}$ can be expanded:

1. with respect to the vectors of one basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ via

$$\mathbf{A} = A^1 \mathbf{e}_1 + A^2 \mathbf{e}_2 + A^3 \mathbf{e}_3 = A^i \mathbf{e}_i.$$

2. with respect to the vectors of the dual basis $\mathbf{e}^1, \mathbf{e}^2, \mathbf{e}^3$ via

$$\mathbf{A} = A_1 \mathbf{e}^1 + A_2 \mathbf{e}^2 + A_3 \mathbf{e}^3 = A_i \mathbf{e}^i.$$

$A^i = \mathbf{A} \cdot \mathbf{e}^i$ are called the **contravariant** components of $\mathbf{A}$,
whereas

$A_i = \mathbf{A} \cdot \mathbf{e}_i$ are called the **covariant** components of $\mathbf{A}$.

RECAP: THE TRANSFORMATION RULE

The transformation rule

The **transformation rule** for the **covariant** components is

$$A'_i = L_{i'}^j A_j.$$

where

$$L_{i'}^j = \mathbf{e}'_i \cdot \mathbf{e}^j,$$

The **transformation rule** for the **contravariant** components is

$$A'^i = L_j^{i'} A^j$$

where

$$L_j^{i'} = \mathbf{e}_j \cdot \mathbf{e}'^{i'}.$$

RECAP: THE RELATIONSHIP BETWEEN COMPONENTS

The relationship between components

Formula to relate covariant and contravariant components:

$$A_i = g_{ik} A^k,$$

$$A^i = g^{ik} A_k,$$

where

$$g_{ik} = g_{ki} = \mathbf{e}_i \cdot \mathbf{e}_k$$

$$g^{ik} = g^{ki} = \mathbf{e}^i \cdot \mathbf{e}^k$$

is called the **metric tensor**.

COVARIANT, CONTRAVARIANT AND MIXED COMPONENTS OF A TENSOR

Covariant and contravariant - first-rank tensors

In dimension 3, a **first-rank** tensor (i.e. a **vector**) $\mathbf{A}$ in a generalised coordinate system is uniquely determined either by

- its three covariant components A_i ,
- its three contravariant components A^i

Covariant and contravariant components are related:

$$\begin{aligned}A_i &= g_{ik} A^k, \\ A^i &= g^{ik} A_k\end{aligned}$$

where g_{ik} and g^{ik} are determined by the basis of the coordinate system .

COMPONENTS OF TENSORS OF HIGHER RANK

Question:

What about tensors of higher ranks?

Tensors of higher rank

Similarly to the rank 1 case, tensors of rank two or higher rank have similar components in a generalised coordinate system:

- We still have the usual **covariant** and **contravariant** components, but we also get some **mixed** ones.
- For example, for a rank two tensor (relaxing the assumption of a Cartesian coordinate system), we have:
 - ▶ **covariant** A_{ik} ,
 - ▶ **contravariant** A^{ik} ,
 - ▶ **mixed** $A_i{}^k$, $A^i{}_k$.

Tensors of higher rank

Similarly to the rank 1 case:

- Covariant, contravariant, and mixed components transform differently under changes of basis.
 - ▶ These transformations are described using transformation matrices $L_{i'}^k$ and $L_k^{i'}$.
- There are formulae relating the three kinds of components.
 - ▶ These relationships are expressed using metric tensors g_{ik} and g^{ik} .

Next, we will reconsider second-rank tensors, relaxing the assumption of a Cartesian coordinate system.

SECOND-RANK TENSORS - COVARIANT AND CONTRAVARIANT COMPONENTS

Second-rank tensors - covariant and contravariant components

In dimension 3, a second rank tensor is specified by its **nine** components, which can be

- **covariant** A_{ik} ,
- **contravariant** A^{ik} ,
- **mixed** $A_i^{\cdot k}$, $A^i_{\cdot k}$.

SECOND-RANK TENSORS - NOTATION OF MIXED COMPONENTS

Notation

Notice that the mixed components are denoted with dots:

$$A_i^{\cdot k} \quad \text{and} \quad A_{\cdot k}^i.$$

The dot in the mixed components emphasises the order of occurrence of the indices.

For instance, in $A_i^{\cdot k}$, the first index is “covariant” and the second “contravariant”:

$$“A_i^{\cdot k} = (A_i)^k”$$

while in $A_{\cdot k}^i$, the first index is “contravariant” and the “second covariant”:

$$“A_{\cdot k}^i = (A^i)_k”.$$

Second-rank tensors - transformations

The components are transformed via:

$$A'_{ik} = L_{i'}^{\ell} L_{k'}^m A_{\ell m},$$

$$A'^{ik} = L_{\ell}^{i'} L_m^{k'} A^{\ell m},$$

$$A_i'^{\cdot k} = L_{i'}^{\ell} L_m^{k'} A_{\ell}^{\cdot m},$$

$$A_{\cdot k}^i = L_{\ell}^{i'} L_{k'}^m A_{\cdot m}^{\ell},$$

where $L_{i'}^k$ and $L_i^{k'}$ ($i, k = 1, 2, 3$) are the coefficients of the direct and the inverse transformations.

SECOND-RANK TENSORS - RELATIONS BETWEEN COMPONENTS

Second-rank tensors - relations between components

The relations between the various components of a tensor are:

$$\begin{aligned}A_{ik} &= g_{il}g_{km}A^{\ell m} = g_{kl}A_i^{\cdot\ell} = g_{il}A_{\cdot k}^{\ell}, \\A^{ik} &= g^{il}g^{km}A_{\ell m} = g^{il}A_{\ell}^{\cdot k} = g^{kl}A_{\cdot\ell}^i, \\A_i^{\cdot k} &= g^{kl}A_{il} = g_{il}A^{\ell k}, \\A_{\cdot k}^i &= g^{il}A_{\ell k} = g_{kl}A^{i\ell}.\end{aligned}$$

SECOND-RANK TENSORS – SYMMETRY AND ANTISYMMETRY

Second-rank tensors – Symmetry and Antisymmetry

Symmetry and antisymmetry apply only to pairs of indices **in the same positions**:

- $A_{ik}^{\cdot\cdot\ell}$ is **symmetric** in i and k if

$$A_{ik}^{\cdot\cdot\ell} = A_{ki}^{\cdot\cdot\ell}.$$

- $B_{\cdot\cdot\ell}^{ik}$ is **anti-symmetric** in i and k if

$$B_{\cdot\cdot\ell}^{ik} = -B_{\cdot\cdot\ell}^{ki}.$$

EXAMPLE 1

Example 1

Let B^k be an arbitrary **contravariant vector**.

We will show that if

$$A_{ijk}B^k$$

is a **second-rank covariant tensor**, then

$$A_{ijk}$$

is a third-rank covariant tensor.

EXAMPLE 1 - PART 2

Example 1

We must show that A_{ijk} is a third-rank covariant tensor. That is

$$A'_{ijk} = L_{i'}^{\ell} L_{j'}^m L_{k'}^n A_{\ell mn}.$$

Info we have:

- $A_{ijk} B^k$ is a second-rank tensor, that is

$$A'_{ijk} B'^k = L_{i'}^{\ell} L_{j'}^m A_{\ell mn} B^n.$$

- B^k is a vector, thus

$$B'^k = L_n^{k'} B^n, \quad \text{or} \quad B^n = L_{k'}^n B'^k.$$

Thus, we have

$$A'_{ijk} B'^k = L_{i'}^{\ell} L_{j'}^m A_{\ell mn} L_{k'}^n B'^k.$$

EXAMPLE 1 - PART 3

Example 1

So far, we have:

$$A'_{ijk}B'^k = L_{i'}^{\ell}L_{j'}^m A_{\ell mn}L_{k'}^n B'^k.$$

Because B^k is arbitrary, we can assume $B'^k \neq 0$. We then cancel it out in the expression above and get

$$A'_{ijk} = L_{i'}^{\ell}L_{j'}^m L_{k'}^n A_{\ell mn},$$

as desired.

EXAMPLE 2

Example 2

Suppose A_i is a covariant tensor and B^i is a contravariant tensor.

Prove that $A_i + B^i$ does not form a tensor.

Solution: Info we have:

- A_i is a covariant tensor, that is $A'_i = L_{i'}^k A_k$,
- B^i is a contravariant tensor, that is $B'^i = L_k^{i'} B^k$.

Hence

$$A'_i + B'^i = L_{i'}^k A_k + L_k^{i'} B^k,$$

which is not a tensor transformation law.

(Transformation law should be:

$$(A_k + B^k)' = A'_k + B'^k = L_{i'}^k (A_i + B^i) \quad (\text{or } L_k^{i'} (A_i + B^i)).$$

PRACTICAL QUESTION

Your turn!

- Let A_{ikl} be a covariant tensor of order 3,
- and B^{pqmn} a contravariant tensor of order 4.

Prove that

$$A_{ikl}B^{klmn}$$

is a mixed tensor of order 3 (with one covariant and two contravariant indices).

Note: In generalised coordinates, the orthogonality property of the rotation matrix is

$$L_{k'}^q L_s^{k'} = \delta_s^q.$$

EXAMPLE 3

Example 3

Given a Cartesian coordinate system K with orthonormal basis $\mathbf{i}_1, \mathbf{i}_2, \mathbf{i}_3$, consider the second-order tensor with components

$$[A_{ik}] = [A^{ik}] = [A_i^{\cdot k}] = [A^i_{\cdot k}] = \begin{pmatrix} 2 & 1 & 3 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{pmatrix}.$$

Let K' be a new coordinate system with basis vectors

$$\mathbf{e}_1 = \mathbf{i}_1,$$

$$\mathbf{e}_2 = \mathbf{i}_1 + \mathbf{i}_2,$$

$$\mathbf{e}_3 = \mathbf{i}_1 + \mathbf{i}_2 + \mathbf{i}_3.$$

Express the covariant, contravariant and mixed components of the given tensor in the system K' .

EXAMPLE 3 - MATRIX OF CHANGE OF COORDINATES

Example 3 - Matrix of change of coordinates

To express the covariant components in terms of K' , we use

$$A'_{ik} = L_{i'}^{\ell} L_{k'}^m A_{\ell m},$$

where $L_{i'}^{\ell}$, $L_{k'}^m$ define how you transform from one basis to another. We need to find the $L_{i'}^{\ell}$. Since

$$\mathbf{e}_1 = \mathbf{i}_1,$$

$$\mathbf{e}_2 = \mathbf{i}_1 + \mathbf{i}_2,$$

$$\mathbf{e}_3 = \mathbf{i}_1 + \mathbf{i}_2 + \mathbf{i}_3.$$

we see that

$$\begin{pmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \mathbf{e}_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{i}_1 \\ \mathbf{i}_2 \\ \mathbf{i}_3 \end{pmatrix}.$$

EXAMPLE 3 - THE MATRIX L

Example 3 - The matrix L

Because

$$\begin{pmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \\ \mathbf{e}_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{i}_1 \\ \mathbf{i}_2 \\ \mathbf{i}_3 \end{pmatrix}.$$

we obtain

$$\begin{aligned} L_{1'}^1 &= 1, & L_{1'}^2 &= 0, & L_{1'}^3 &= 0, \\ L_{2'}^1 &= 1, & L_{2'}^2 &= 1, & L_{2'}^3 &= 0, \\ L_{3'}^1 &= 1, & L_{3'}^2 &= 1, & L_{3'}^3 &= 1. \end{aligned}$$

EXAMPLE 3 - COVARIANT COMPONENTS

Example 3 - Covariant components

Now we only need to apply the transformation law

$$A'_{ik} = L_{i'}^{\ell} L_{k'}^m A_{\ell m}.$$

Notice that

$$A'_{11} = L_{1'}^{\ell} L_{1'}^m A_{\ell m} = \sum_i \sum_k L_{i'}^{\ell} L_{k'}^m A_{\ell m}.$$

Thus, computing this for all i, k is too much work. Instead, notice that, if we set $L = [L_{i'}^j]$ and $A = [A_{ij}]$, then

$$A'_{ik} = [LAL^T]_{ik}.$$

(See this week's exercises for details.)

Example 3 - Covariant components

Thus, we only need to perform matrix multiplication:

$$\begin{aligned} A' &= [A'_{ik}] = LAL^T \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 3 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 3 & 6 \\ 4 & 8 & 15 \\ 5 & 11 & 19 \end{pmatrix}. \end{aligned}$$

EXAMPLE 3 -CONTRAVARIANT AND MIXED COMPONENTS

Example 3 -Contravariant and Mixed Components

To find A'^{ik} , $A'_i{}^k$ and $A'^i{}_k$, we use the formulae

$$A'^{ik} = g^{il} g^{km} A'_{\ell m},$$

$$A'_i{}^k = g^{k\ell} A'_{i\ell},$$

$$A'^i{}_k = g^{i\ell} A'_{\ell k},$$

where $G = [g^{ik}] = [\mathbf{e}^i \cdot \mathbf{e}^k] = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$

EXAMPLE 3 - CONTRAVARIANT AND MIXED COMPONENTS - PART 2

Example 3 -Contravariant and Mixed Components

So, to find the contravariant components, we can show that it suffices to compute

$$\begin{aligned}[A'^{ik}] &= GA'G^T, \\ [A'_i{}^k] &= A'G^T = A'G, \quad (G \text{ is symmetric}), \\ [A'^i{}_k] &= GA'.\end{aligned}$$

PRACTICAL QUESTION

Your turn!

Given a Cartesian coordinate system K with orthonormal basis $\mathbf{i}_1, \mathbf{i}_2, \mathbf{i}_3$, consider the second-order tensor with components

$$\mathbf{B} = [B_{ik}] = [B^{ik}] = [B_i^{\cdot k}] = [B^i_{\cdot k}] = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Let K' be a new coordinate system with basis vectors

$$\mathbf{e}_1 = \mathbf{i}_1 - \mathbf{i}_2,$$

$$\mathbf{e}_2 = \mathbf{i}_1 + 4\mathbf{i}_2 + \mathbf{i}_3,$$

$$\mathbf{e}_3 = \mathbf{i}_1 + 3\mathbf{i}_2 + \mathbf{i}_3.$$

1. Compute the covariant components of $\mathbf{B}$ in the system K' .
2. Compute the dual basis vectors $\mathbf{e}^1, \mathbf{e}^2, \mathbf{e}^3$.
3. Using the metric tensor, compute contravariant components.

Next time...

- Chapter 5: Local Coordinate Transform
 - ▶ Associated tensors,
 - ▶ The metric tensor,
 - ▶ Higher order tensors in generalised coordinates.